PAPER POSTER CODE

Causal Discovery in Mixtures of Populations

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Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI 2026) • PMLR 337:4457–4478

1. Pooling Populations Confounds Everything

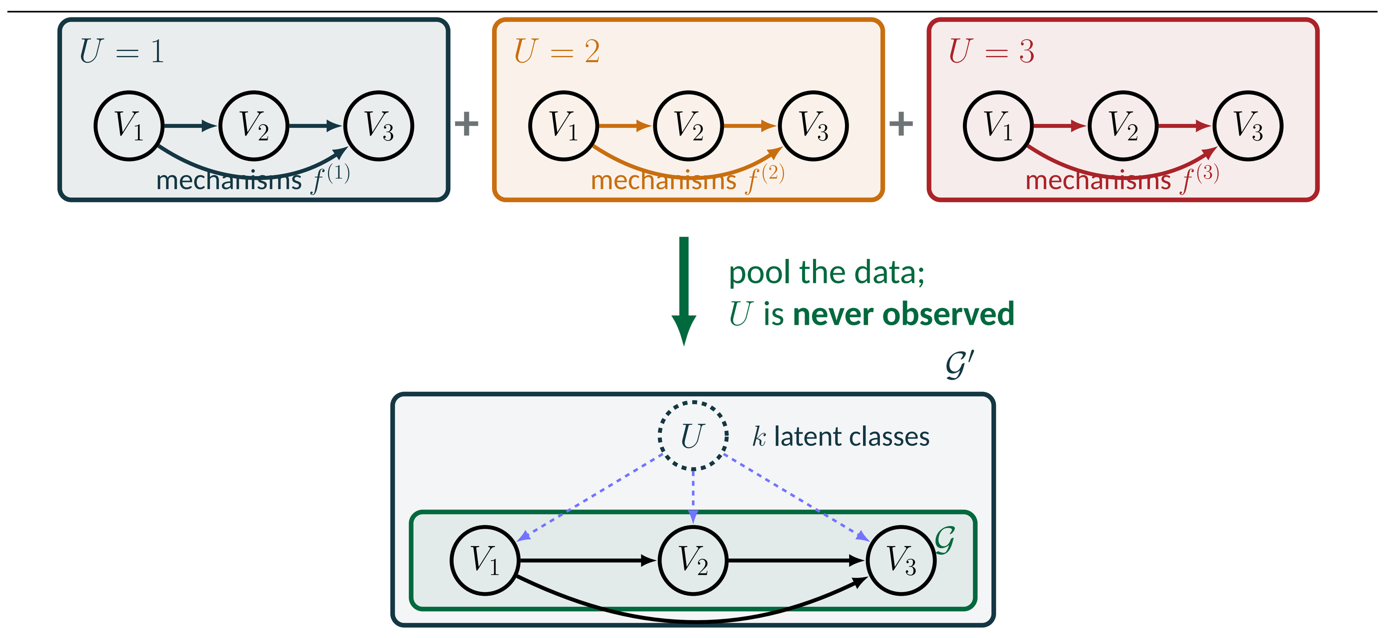
The problem, and why the standard toolkit fails

Goal. Recover the observed sub-graph $\mathcal{G}$ up to its MEC (a CP-DAG) from the pooled marginal $\Pr(\mathbf{V})$ alone, never observing U.

- U sits in *every* separating set, so **no conditional independence survives.** PC returns the complete graph.
- FCI assumes a confounder touches only *two* observables, and returns coarser classes than an MEC. [13,14]
- Pervasive-confounding methods buy identifiability with *parametric* assumptions: linear + sub-Gaussian [6], non-Gaussian cumulants [4], finite-dimensional Hilbert space [1].
- A *discrete* universal confounder: open until now.

Assumptions: faithfulness w.r.t. $\mathcal{G}'$; U discrete with *known* k; no other latents; Bernoulli observables — the hardest case — with **arbitrary** structural equations and noise.

One DAG, k mechanisms



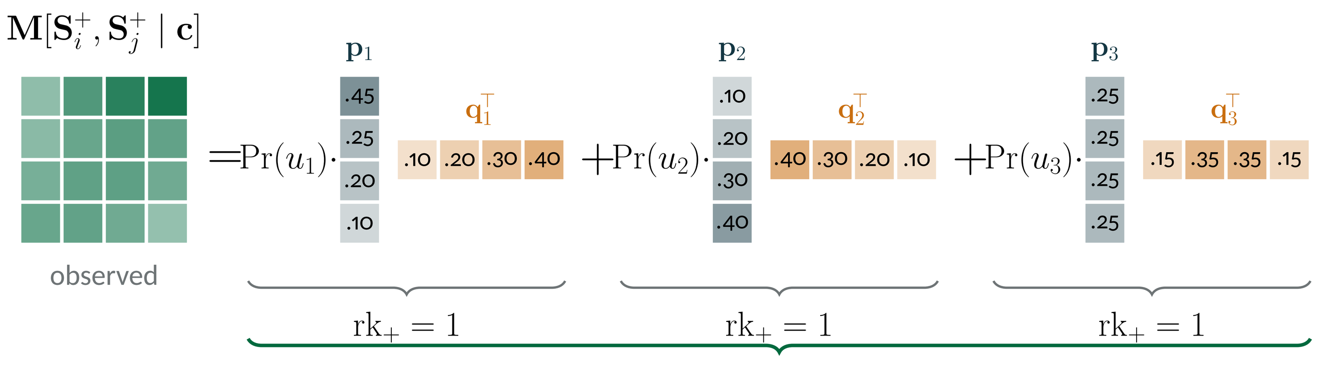
every pair of observables is d-connected through U

Contributions

- **First** identifiability result and algorithm for DAG discovery under mixture confounding, with **no** parametric assumptions on the observables.
- **Rank tests** replace CI tests — CI testing is the $k=1$ case of a family indexed by the complexity of the confounding.
- A calibrated **hypothesis test for matrix rank**, reusable in latent-variable models and causal representation learning [15].
- **Agglomeration** of low-cardinality variables, plus theory to pull group conclusions back to single vertices.

2. Rank Replaces Conditional Independence

Read the joint distribution as a matrix, $\mathbf{M}[\mathbf{S}_i^+, \mathbf{S}_j^+ \mid \mathbf{c}]_{x,y} := \Pr(x, y \mid \mathbf{c})$. Conditioning on the source as well makes the groups independent, so every summand of $\mathbf{M} = \sum_u \Pr(u)\, \mathbf{p}_u \mathbf{q}_u^\top$ is an outer product:



$$\mathrm{rk}_+\big(\mathbf{M}[\mathbf{S}_i^+, \mathbf{S}_j^+ \mid \mathbf{c}]\big) \le k$$ (drawn for $k=3$)

with $\mathbf{p}_u = \Pr(\mathbf{S}_i^+ \mid u, \mathbf{c})$, $\mathbf{q}_u = \Pr(\mathbf{S}_j^+ \mid u, \mathbf{c})$: sixteen observed entries generated by k pairs of vectors.

Rank Test (Lemma 1)

For V_i, V_j of cardinality $> k$ confounded by U in $\mathcal{G}'$,

$$V_i \perp\!\!\!\perp_d^{\mathcal{G}} V_j \mid \mathbf{C} \iff \mathrm{rk}_+(\mathbf{M}[V_i, V_j \mid \mathbf{C}]) \le k$$

generically — exceptions have Lebesgue measure 0.

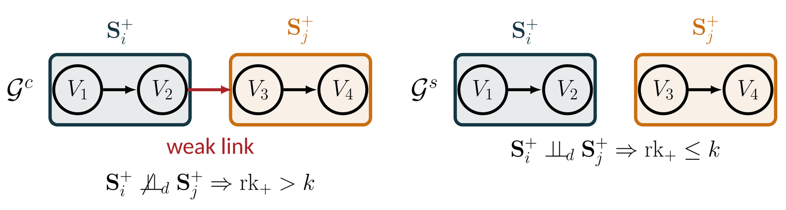
Both directions, and a sanity check

- $\le$: d-separation gives k rank-one summands, at *any* cardinality.
- $>$: faithfulness summand-wise. Each new $\mathbf{p}_u$ lands in the span of its predecessors only on a measure-zero set, so the ranks add; cardinality $> k$ leaves room for them to.
- $k=1$: $\mathrm{rk}_+ \le 1$ *is* independence. A CI test is a rank test with nothing to deconfound.

3. Testing the Rank of a Matrix of Probabilities

Why a real test is needed

- With finite samples $\sigma_{k+1}(\hat{\mathbf{A}}) > 0$ *always*; rank is never exactly deficient.
- Thresholding σ_{k+1} [3] needs a cutoff unavailable *a priori* and carries no significance level.
- We run *many* rank tests per candidate pair, so we need p-values, not magnitudes.



The models differ *only* by the weak link $V_2 \to V_3$: the hardest possible discrimination.

The statistic

Write $\hat{\mathbf{A}} = \mathbf{A} + \varepsilon(N)\mathbf{B}$, $\varepsilon(N) \propto 1/\sqrt{N}$ [12]. From $\hat{\mathbf{A}} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ keep the *trailing* singular vectors $\mathbf{U}_2, \mathbf{V}_2$; put $\hat{l} = \mathrm{vec}(\mathbf{U}_2^\top \hat{\mathbf{A}} \mathbf{V}_2)$ and $\hat{\mathbf{Q}} = (\mathbf{V}_2^\top \otimes \mathbf{U}_2^\top)\hat{\Sigma}(\mathbf{V}_2 \otimes \mathbf{U}_2)$ with $\hat{\Sigma}$ the Bernoulli cell-count covariance. Then

$$N\, \hat{l}^\top \hat{\mathbf{Q}}^\dagger \hat{l} \xrightarrow{d} \chi_f^2, \quad f = (m-k)(n-k),$$

pseudoinverting *after* the projection, not before. Equivalently $f = \mathrm{rk}_+(\hat{\mathbf{Q}})$: it counts the "extra" singular directions under test.

Sanity check. At $k=1$ this gives $f = (m-1)(n-1)$ — exactly the degrees of freedom of Pearson's χ^2 test of independence on an $m \times n$ contingency table.

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pip install probrank
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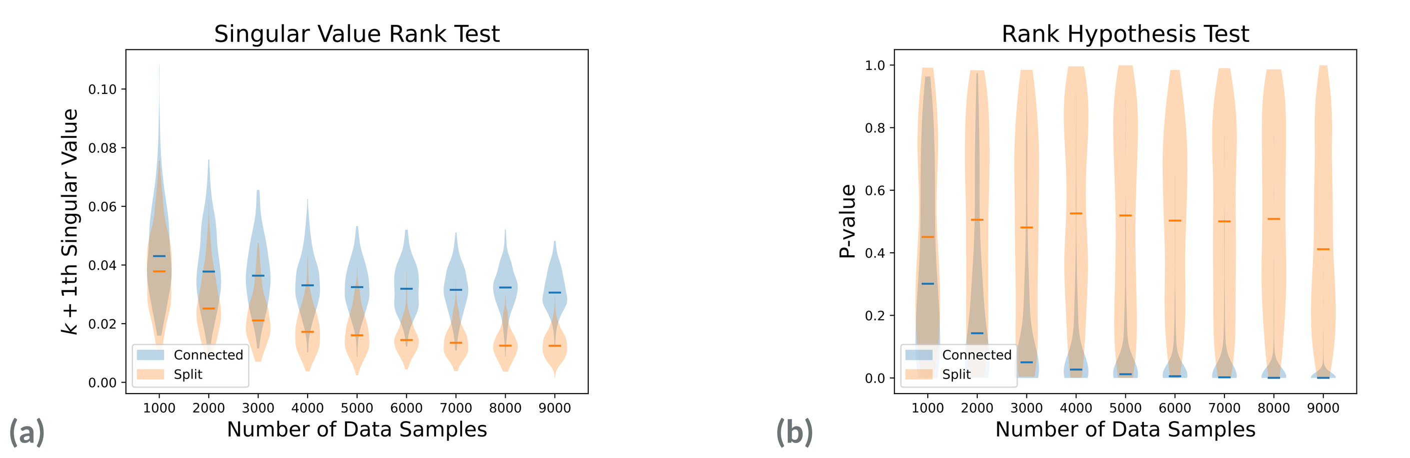
Empirical test



200 runs, $k=2$, 1000–9000 samples. (a) the $(k{+}1)$th singular values overlap and admit no natural cut; (b) p-values separate cleanly past ~ 2000 samples *and* carry a calibrated significance level.

4. Coarsening: Agglomerate Variables into Supervariables

The cardinality problem

Bernoulli observables give a 2×2 matrix, whose rank never exceeds 2: for $k \ge 2$ the rank test is **vacuous. Fix:** agglomerate $\lceil \lg(k+1) \rceil$ binary variables into a supervariable of cardinality $2^{\lceil \lg(k+1) \rceil} \ge k+1$.

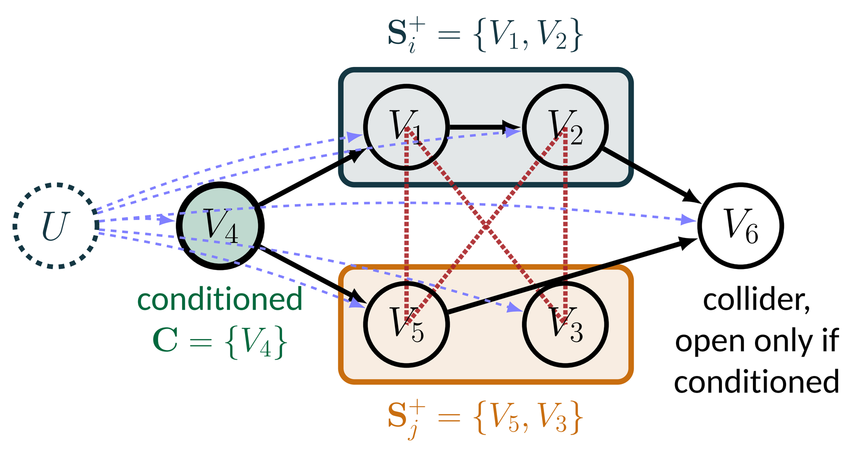
Independence-Preserving Agglomeration

$(\mathbf{S}_i \cup \{V_i\},\ \mathbf{S}_j \cup \{V_j\})$ is an **IPA** of (V_i, V_j) if $\mathbf{S}_i^+ \perp\!\!\!\perp_d^{\mathcal{G}} \mathbf{S}_j^+ \mid \mathbf{C}$ for some $\mathbf{C}$.

Worked example ($k=2$)

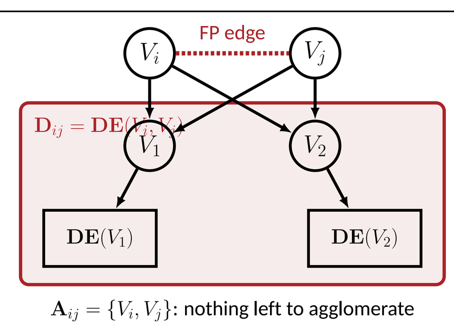
$V_1 \perp\!\!\!\perp_d V_5 \mid \{V_4\}$: the fork at V_4 is blocked and the path through V_6 dies at an unconditioned collider. Agglomerating $\{V_1, V_2\}$ against $\{V_5, V_3\}$ is an IPA, so $\mathbf{M}[\{V_1,V_2\},\{V_5,V_3\} \mid v_4]$ is 4×4 with $\mathrm{rk}_+ \le 2$ — deleting **all four** cross-group non-edges at once while leaving the within-group edge $V_1 - V_2$ alone. The 2×2 matrix $\mathbf{M}[V_1, V_5 \mid v_4]$ says nothing at all.

Phase I: PC's edge-removal loop over *pairs of agglomerations* rather than pairs of vertices, rank tests for CI tests — $|\mathbf{V}|^{\mathcal{O}(\Delta^2 \log k)}$ tests, Δ the max degree.



5. When Coarsening Fails, and How k-MixProd Repairs It

False positive edges



$\mathbf{A}_{ij} = \{V_i, V_j\}$: nothing left to agglomerate

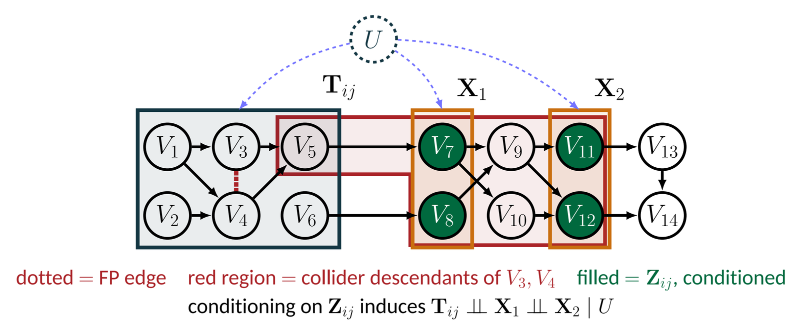
Conditioning on a common descendant *opens* the collider $V_i \to \cdot \leftarrow V_j$, so they can be neither conditioned on nor agglomerated. When almost every other vertex is one, no IPA exists and a **false-positive edge** survives Phase I.

FP edges are confined

Early vertices $\mathbf{H} := \{V : |\overline{\mathbf{DE}}(V)| < (2+\Delta^2)\lceil \lg(k+1) \rceil - 2\}$ satisfy $|\mathbf{H}| \le (2+\Delta^2)\lceil \lg(k+1) \rceil - 2$ and $\mathbf{E}_1 \setminus \mathbf{E} \subseteq \mathbf{H} \times \mathbf{H}$.

Phase II: refine inside a k-MixProd instance [7,8]

The within-source $\Pr(\mathbf{V} \mid u)$ has *no* global confounder, so ordinary CI tests work there — recover it on a small subset. $\mathbf{T}_{ij} = \{V_i, V_j\} \cup \mathbf{NB}_1^{\mathcal{G}_1}(V_i, V_j)$ is guaranteed to contain a separating set. Pick $\mathbf{X}_1, \mathbf{X}_2$ far away, condition on $\mathbf{Z}_{ij} = \mathbf{NB}_2(\mathbf{X}_1) \cup \mathbf{NB}_2(\mathbf{X}_2)$, deconfound, then test *within each class*: drop the edge iff $V_i \perp\!\!\!\perp V_j \mid \mathbf{C}, u$ for all u. **Phase III** = immoralities + Meek rules, exactly as PC.



dotted = FP edge red region = collider descendants of V_3, V_4 filled = $\mathbf{Z}_{ij}$, conditioned
conditioning on $\mathbf{Z}_{ij}$ induces $\mathbf{T}_{ij} \perp\!\!\!\perp \mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2 \mid U$

k-MixProd identifiability [2,9]

X_1, X_2, X_3 of cardinality $\kappa_1, \kappa_2, \kappa_3$, mutually independent given $U \in \{1, \ldots, k\}$, are generically identifiable if $\min(\kappa_1, k) + \min(\kappa_2, k) + \min(\kappa_3, k) \ge 2k+2$ — so Phase I need only expose enough sparsity to d-separate three large enough groups.

Cost

With oracles of cost ρ (non-negative rank) and τ (k-MixProd), and Phase II calling the latter $\mathcal{O}(k|\mathbf{E}|2^{\Delta^2})$ times, the algorithm runs in

$$|\mathbf{V}|^{\mathcal{O}(\Delta^2 \log k)}\rho + \mathcal{O}\big((\Delta^2 \lg k + |\mathbf{E}|)k2^{\Delta^2}\big)\tau.$$

ρ: the non-negative rank of a $(k{+}1) \times (k{+}1)$ matrix costs $k^{\mathcal{O}(k^2)}$ [10], but ordinary rank tests substitute well in practice [3] — and ours is $\mathcal{O}(k^6)$, the inversion of a $(k{+}1)^2 \times (k{+}1)^2$ covariance.

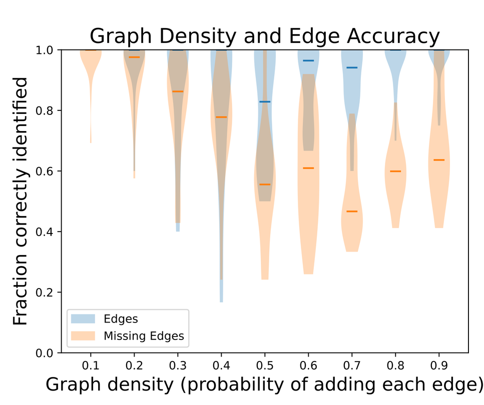
τ: k-MixProd is an order-3 tensor decomposition of size $\mathcal{O}(k)^3$, solvable in $\mathcal{O}(k^{6.05})$ when the rank is linear in k [5] — though the problem is unstable, with worst-case sample complexity exponential in k [11].

In practice Phase II is optional: it only repairs FP edges, and those are confined to $\mathbf{H} \times \mathbf{H}$.

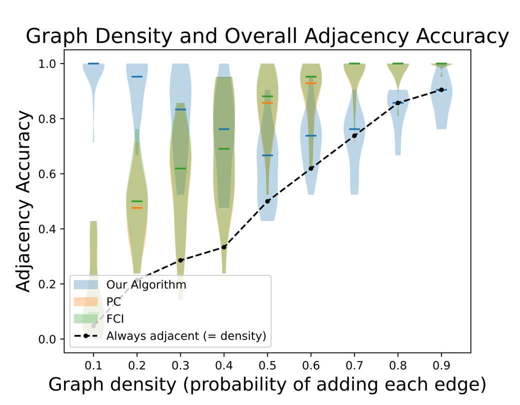
Identifiability (Corollary 1)

If $\mathcal{G} = (\mathbf{V}, \mathbf{E})$ has degree bound Δ and mixture source $U \in \{1, \ldots, k\}$ with $|\mathbf{V}| \ge (\Delta^3 + 2\Delta^2 + 4\Delta + 2)\lceil \lg(k+1) \rceil + 2\Delta^2 + 2\Delta^3$, then $\mathcal{G}$ is **generically identifiable up to its MEC**, with no parametric assumptions on the observables. The bound is loose — identifiability arrives far sooner in practice — but it is the first and only such guarantee.

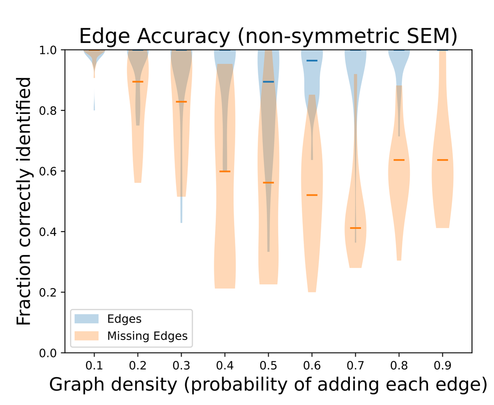
6. Empirical Performance



(a) Strongest in the **sparse regime** the theory targets; at higher density we start dropping true edges and keeping spurious ones.



(b) PC and FCI [16] ignore U and return roughly constant moderate density (14–20 of 21 edges) whatever the truth — barely above the "always adjacent" baseline (dashed).



(c) A non-symmetric weighted-logistic SEM — non-additive, sensitive to *which* parents fire — leaves the algorithm untouched and the curves comparable.

Setup and takeaways

Random Erdős–Rényi DAGs on 7 vertices, edge probability $.1$–$.9$, 20 graphs each, $10{,}000$ samples, $k=2$, binary observables.

Our method leads exactly where PC and FCI are at chance.

Misspecified k is easy to spot: too small gives a near-complete graph, too large a near-empty one — so k can be tuned until the output is neither.

Open questions. Every candidate pair undergoes many rank tests — over agglomerations, conditioning sets, and their assignments — so a fixed per-test level is not a fixed family-wise level. A principled multiplicity correction, and the *necessity* of the identifiability bound, remain open.

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Acknowledgements

B. M. is partially supported by the Advanced Research Concepts (ARC) COMPASS program, sponsored by DARPA under agreement HR001-25-3-0212. Y. R. is partially supported by BSF grant 2023607. L. S. is partially supported by NSF grant CCF-2321079.